

Deep Residual Learning for Cross-Source Ground-Based Cloud Classification: Grouped Train/Test Splitting, Label Matching, and Training Behavior

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Abstract

Ground-based sky imaging provides high-spatiotemporal resolution observations of cloud fields, supporting solar irradiance forecasting, aviation meteorology, and climate modeling. However, automated classification models deployed across disparate ground sensors encounter domain shift from different camera views (regional cloud views versus wide-angle whole-sky cameras) and different labeling schemes between fine-grained World Meteorological Organization (WMO) genera and operational sky-condition categories. In this project, we address these challenges across the deep residual learning family (ResNet-18, ResNet-34, and ResNet-50). First, an exact-byte duplicate audit identifies cross-split contamination in public benchmark releases (156 duplicate clusters across official GCD splits, and 3 conflicting pairs in CCSN). We apply grouped train/test splitting via Stratified Group K -Fold to keep duplicate images in the same split, defining an 80% development pool and an untouched 20% final test set. Second, we define five shared cloud classes mapping fine-grained WMO genera and operational sky conditions into a common label space, excluding clear sky, contrails, and mixed scenes to focus on single-cloud categories. Third, to address the dataset size imbalance (85.96% GCD versus 14.04% CCSN), we balance training by sampling both datasets equally on average per batch (balanced sampling), supported by hyperparameter exploration on the development pool. On the held-out test set, the joint model achieves a source-balanced accuracy of **74.58%** on ResNet-18, **74.52%** on ResNet-34, and **74.66%** on ResNet-50. Because the gap between ResNet-18 and ResNet-50 is only 0.08 percentage points, ResNet-18 serves as a practical, lightweight baseline for resource-constrained deployment. Furthermore, joint training improves classification accuracy on the sparse CCSN test component by +4.06 percentage points on ResNet-18 (56.84% to 60.90%), while cross-source evaluation reveals marked transfer asymmetry between regional and whole-sky modalities.

1 Introduction

Clouds represent the single largest source of uncertainty in global atmospheric radiative transfer, earth energy budget calculations, and numerical weather prediction models [1]. Accurate, real-time quantification of cloud types and vertical structure governs both surface solar irradiance attenuation and downward longwave greenhouse counter-radiation. While geostationary and low-Earth-orbit satellites provide synoptic continental coverage, satellite instruments suffer from coarse surface spatial resolution, sub-pixel cloud fraction ambiguity, parallax errors, and signal attenuation when observing thin boundary layer clouds underneath high cirrus decks [2].

Consequently, ground-based sky imagers have emerged as vital instruments for surface-level atmospheric observation. Ground imagers observe cloud systems directly from the surface, capturing key visual features: cloud base morphology, texture, and convective geometry. Historically,

cloud classification relied upon manual human synoptic observations conducted every three to six hours using World Meteorological Organization (WMO) International Cloud Atlas guidelines [3]. However, human observations remain subjective, limited in temporal frequency, and difficult to scale across remote meteorological stations.

In recent years, deep convolutional neural networks (CNNs) have demonstrated promising capability for automated ground-based cloud classification [4, 5]. Nonetheless, testing across different datasets encounters three practical challenges:

1. **Data Leakage via Duplicate Exposures:** Ground-based sky cameras operating on automated schedules frequently capture identical images under calm atmospheric conditions. Random splitting can distribute identical files across training and evaluation splits, creating a risk of the model memorizing test images and inflating evaluation metrics.
2. **Different Labeling Schemes:** Open-access datasets follow different labeling schemes. The Cirrus Cumulus Stratus Nimbus (CCSN) dataset [4] adheres to WMO genera, whereas the Ground-based Cloud Dataset (GCD) [5] uses operational sky-condition classes. Direct concatenation without matching the labels leads to inconsistent supervision.
3. **Cross-Source Optical Discrepancy:** Sensor hardware varies across observation archives. CCSN images focus on localized regional cloud fields, whereas GCD captures wide-angle whole-sky views, introducing perspective, illumination, and scale domain shifts.

To address these challenges, we compare ResNet-18, ResNet-34, and ResNet-50 [6] using grouped train/test splitting, five shared cloud classes, and balanced batch sampling. Supporting hyperparameter sweeps on the development pool guide model configuration, and final benchmarks are evaluated on an untouched final test set with non-parametric bootstrap confidence intervals. Across sensor modalities, the joint model yields a source-balanced accuracy of 74.58% on ResNet-18 and 74.66% on ResNet-50, supporting ResNet-18 as a practical baseline.

2 Data Sources, Duplicate Removal, and Grouped Train/Test Splitting

2.1 Dataset Profiles: CCSN and GCD

To investigate cross-source domain shift, we examine two primary ground-based benchmarks:

Cirrus Cumulus Stratus Nimbus (CCSN) Dataset: Published by Zhang et al. [4], CCSN contains 2,543 raw images captured in China using digital cameras focusing on specific cloud fields. Images possess diverse resolutions and are categorized into 11 fine-grained classes: 10 WMO genera (Cirrus [Ci], Cirrocumulus [Cc], Cirrostratus [Cs], Altocumulus [Ac], Altostratus [As], Stratocumulus [Sc], Stratus [St], Cumulus [Cu], Cumulonimbus [Cb], Nimbostratus [Ns]) and one human-generated category (Contrails [Ct], 200 images). Three conflicting duplicate pairs (6 images) with identical SHA-256 file-byte hashes but conflicting genus labels were purged, leaving 2,537 verified images in the 11-class view, and 2,337 images in the 5-class harmonized benchmark after contrail exclusion.

Ground-based Cloud Dataset (GCD): Published by Liu et al. [5], GCD comprises 19,000 images collected by ground-based all-sky cameras across nine Chinese provinces during 2019–2020. Images capture wide whole-sky perspectives categorized into seven operational sky classes: 1_cumulus, 2_altocumulus, 3_cirrus, 4_clearsky, 5_stratocumulus, 6_cumulonimbus, and 7_mixed. To construct a consistent cloud-type classification benchmark, we exclude 955 mixed-sky scenes (7_mixed) and 3,739 clear-sky scenes (4_clearsky, which permits $\leq 10\%$ cloudiness per GCD authors), yielding 14,306 cloud images in the 5-class harmonized benchmark.

2.2 Observational Redundancy and Exact-Byte Duplicate Audit

In automated sky imaging systems, consecutive exposures under calm conditions frequently record byte-identical image files. An exact-byte SHA-256 file hash audit of the public GCD release identified 156 duplicate clusters spanning the official train and test directories, primarily in clear-sky and static sky conditions. Evaluating models on official splits containing cross-split duplicates risks the model memorizing test images.

Furthermore, in CCSN, the audit identified three conflicting duplicate pairs (six images with identical SHA-256 file-byte hashes) assigned conflicting genus labels: two pairs conflicting between Altocumulus (*Ac*) and Altostratus (*As*), and one pair conflicting between Cirrocumulus (*Cc*) and Cirrostratus (*Cs*). These six images were removed to maintain label consistency.

2.3 Splitting Protocol and Sample Allocation

To maintain benchmark reliability, we applied consistent data curation and splitting rules:

1. **Purging Conflicting Duplicates:** The three conflicting duplicate pairs in CCSN were removed.
2. **Exclusion of Contrails and Mixed Scenes:** Anthropogenic contrails (*Ct* in CCSN, 200 images) and heterogeneous mixed scenes (*7_mixed* in GCD, 955 images) were excluded to focus on natural single-cloud scenes.
3. **Exclusion of Clear Sky (*4_clearsky*):** GCD contains 3,739 clear-sky images, whereas CCSN contains zero clear-sky images. Including clear sky would create an extreme domain disparity and risk shortcut learning on blue sky color; thus *4_clearsky* was excluded.
4. **Keeping Duplicate Groups Intact via Stratified Group K -Fold:** Exact-byte duplicate clusters share a common `group_id`. All splits were generated using `StratifiedGroupKFold` (seed 42), keeping each duplicate group intact so identical exposures never cross between train and test splits. (Note that exact-byte grouping does not identify near-duplicate scenes with minor sensor noise).

We define terminology for all data splits as summarized in Table 1:

- **Development Pool ($N = 13,313$ images):** The 80% development pool split into parameter training ($N = 10,650$) and an **internal development validation set** ($N = 2,663$, fold 0). Validation data are used only for hyperparameter exploration and checkpoint selection; they do not constitute final benchmark evidence.
- **Held-Out Test Set ($N = 3,330$ images):** The untouched 20% evaluation set. It decomposes into two source-specific components: the **Harmonized CCSN Test Component** ($N = 468$) and the **Harmonized GCD Test Component** ($N = 2,862$).
- **Pooled Test Set ($N = 3,330$ images):** The sample-concatenated evaluation set. Because GCD accounts for 85.95% of this pool ($2,862/3,330$), pooled combined accuracy is inherently GCD-heavy. Consequently, **source-balanced averaging** is adopted as the primary headline metric, treating both source domains with equal weight.

3 Combining the Two Datasets: A Shared Five-Class Labeling Scheme

3.1 Shared Labeling Scheme

The WMO International Cloud Atlas defines ten cloud genera based on physical altitude and convective structure, whereas ground-based whole-sky imagers typically aggregate fine genera into broader operational categories. We therefore map both datasets into a shared labeling scheme.

Dataset Split / View	Classes	Train	Val (Dev)	Test	Total
CCSN 11-Class View	11	1,622	407	508	2,537
GCD 6-Class View	6	11,548	2,888	3,609	18,045
Harmonized CCSN Component	5	1,495	374	468	2,337
Harmonized GCD Component	5	9,155	2,289	2,862	14,306
Pooled Harmonized Benchmark	5	10,650	2,663	3,330	16,643

Table 1: Dataset splits and sample allocation across parameter training, internal development validation, and final test sets.

3.2 Label Mapping

Table 2 details the mapping between source labels and the five shared classes, codified in `metadata/splits/harmonized_5bin_canonical.json`.

Shared Class	CCSN Genera	GCD Category	Physical Cloud Morphology
Cumulus	Cu	1.cumulus	Detached, dense convective clouds with distinct outlines and flat bases.
Alto cumulus	Ac, Cc	2.altocumulus	Mid-level stratocumuliform rolls (<i>Ac</i>) and high-level ice ripples (<i>Cc</i>).
Cirrus	Ci, Cs	3.cirrus	High-altitude fibrous ice filaments (<i>Ci</i>) and thin veil sheets (<i>Cs</i>).
Stratocumulus	Sc, St, As	5.stratocumulus	Low continuous stratiform decks (<i>Sc, St</i>) and dense non-precipitating sheets (<i>As</i>).
Cumulonimbus	Cb, Ns	6.cumulonimbus	Deep convective towers (<i>Cb</i>) and thick precipitation-bearing layers (<i>Ns</i>).

Table 2: Mapping CCSN and GCD into five shared cloud classes.

4 Model Architecture and Orientation Invariance

4.1 Architecture Overview and Trade-offs

We evaluate three architectures from the deep residual network family [6] spanning different computational capacities:

- **ResNet-18:** 8 BasicBlocks (two 3×3 convolutions per block). Parameter count with custom head is 11,309,637 (~ 11.3 M parameters, 1.82 GFLOPs per 224×224 pass).
- **ResNet-34:** 16 BasicBlocks with deeper feature hierarchy (channel depths 64 to 512). Parameter count is 21,417,797 (~ 21.4 M parameters, 3.66 GFLOPs).
- **ResNet-50:** 16 3-layer Bottleneck blocks (1×1 , 3×3 , 1×1 convolutions) reaching 2048 channels. Parameter count is 24,034,373 (~ 24.0 M parameters, 4.12 GFLOPs).

Residual shortcut connections propagate gradients directly across layers during backpropagation, stabilizing optimization across varying depths while providing a practical trade-off between model size and model capacity.

4.2 Regularized Classification Head

The standard ImageNet 1,000-class fully connected layer is replaced by a regularized classification head structured as follows:

- Dropout($p = 0.3$) applied to pooled backbone features x_{pool} ($d_{\text{in}} = 512$ for ResNet-18/34, 2048 for ResNet-50)
- Linear($d_{\text{in}} \rightarrow 256$) $\rightarrow$ BatchNorm1d(256) $\rightarrow$ GELU
- Dropout($p = 0.2$)
- Linear($256 \rightarrow 5$) producing class logits z

Both dropout layers reside within the replacement head; no dropout is inserted into the residual backbone.

4.3 Orientation and Data Augmentation

We do not flip vertically because clouds have an inherent top and bottom, with flat condensation bases at the bottom and vertical growth toward the top. Inverting them would produce physically unrealistic images. Data augmentation is therefore limited to horizontal flipping ($p = 0.5$), bounded rotation ($\pm 15^\circ$), random resized cropping (scale 0.8 to 1.0), and mild color jitter ($\pm 10\%$).

5 Training Behavior and Hyperparameter Exploration

5.1 Differential Learning Rates and Annealing Schedule

We employ differential learning rates between the pretrained backbone and the initialized classification head: $\eta^{(\text{backbone})} = 5 \times 10^{-5}$ and $\eta^{(\text{head})} = 5 \times 10^{-4}$ (a 1:10 ratio), optimized with AdamW and decoupled weight decay $\lambda = 1 \times 10^{-2}$. Learning rates follow a cosine annealing schedule decaying to $\eta_{\text{min}} = 10^{-6}$ over 15 epochs.

5.2 Batch Sampling and Training Setup

In the harmonized training split, GCD accounts for 85.96% of images ($N_{\text{GCD}} = 9,155$) and CCSN contributes 14.04% ($N_{\text{CCSN}} = 1,495$). Uniform random sampling would allow GCD samples to dominate each batch. To balance gradient updates across domains, joint training applies `WeightedRandomSampler` with inverse source-size weights:

$$w_i = \frac{0.5}{N_{\text{domain}(i)}} \quad (1)$$

This procedure samples both datasets equally on average per batch across iterations (balanced sampling).

Notes on the training setup

- **Image exposure count:** Because batches sample both datasets equally, images from the smaller CCSN dataset appear about 3.6 times per epoch on average (compared to once per epoch in CCSN-only training). The observed test improvements therefore come from the combined training setup—more gradient updates, balanced batches, and shared features.
- **Validation loss weighting:** While training batches are balanced between the two datasets, the validation split is evaluated over all available validation images (374 CCSN and 2,289 GCD). As a result, model checkpoint selection is weighted more heavily toward GCD validation loss ($L_{\text{val}} \approx 0.14L_{\text{CCSN}} + 0.86L_{\text{GCD}}$).

5.3 Hyperparameter Exploration on Development Data

As supporting exploration, we evaluated ten hyperparameter configurations on the internal development validation set (excluding test samples) covering AdamW and SGD with momentum, learning rates, weight decay, label smoothing, and dropout. Figure 1 displays convergence

trajectories for the selected baseline configuration (Trial 08), while complete validation results across all ten screening runs are documented in Appendix Table 4.

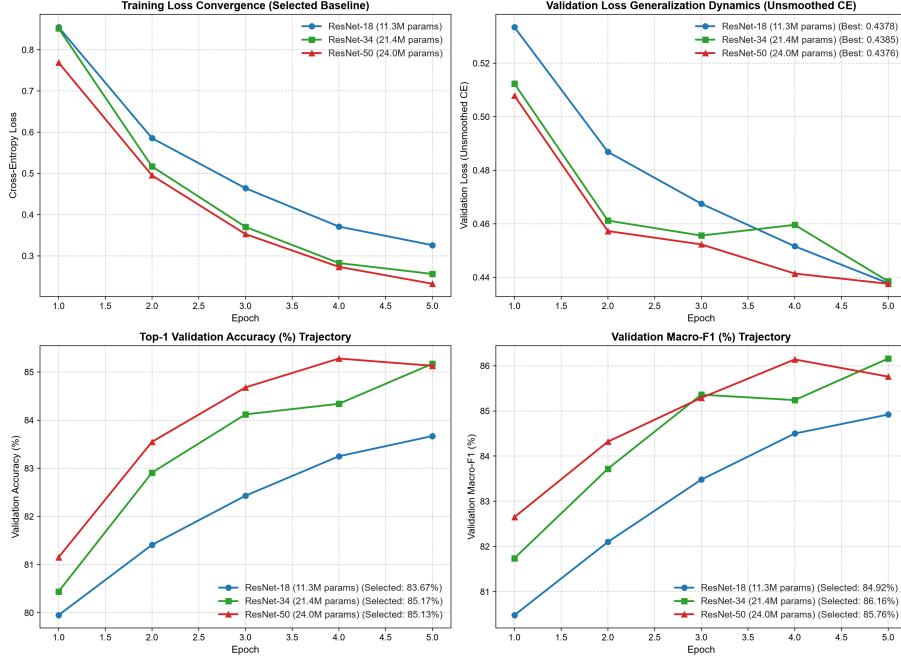


Figure 1: Gradient descent convergence dynamics across ResNet-18, ResNet-34, and ResNet-50 during 5-epoch Trial 08 screening runs on the internal development validation set. Panels display training loss convergence (top-left), unsmoothed validation loss (top-right), validation accuracy (bottom-left), and validation macro-F1 score (bottom-right). Legends indicate checkpoint-selected screening performance.

The screening runs show that differential learning rates with AdamW ($\eta_{bb} = 5 \times 10^{-5}$, $\eta_{head} = 5 \times 10^{-4}$, Trial 08) outperform standard momentum SGD (Trial 04) by 1.2 to 2.0 percentage points across architectures (+1.5 pp for ResNet-18, +2.0 pp for ResNet-34, and +1.2 pp for ResNet-50). The unregularized label smoothing setting ($\alpha = 0.0$) with weight decay $\lambda = 1 \times 10^{-2}$ yielded stable convergence across all three architectures and was selected as the baseline configuration for production benchmarks.

6 Empirical Results and Comparative Production Benchmarks

6.1 Evaluation Protocol

Production models were trained for 15 epochs on the harmonized dataset using the baseline configuration. Checkpoint selection restored the state achieving minimum validation loss, which was subsequently evaluated on the held-out test set (3,330 images). Image-level 95% bootstrap confidence intervals ($B = 1,000$) quantify sampling variability conditionally on the test set.

6.2 Primary Empirical Headline: Source-Balanced Joint Performance

The headline benchmark result of this study is the joint CCSN+GCD model evaluated on the held-out test set, reported with source-balanced averaging across sensor modalities. Table 3 presents this primary comparison.

Several critical observations emerge from Table 3:

1. **Headline Performance and ResNet-18 as Practical Baseline:** ResNet-18 achieves a source-balanced accuracy of **74.58%**, compared to **74.66%** for ResNet-50—a margin of

Architecture	Params	CCSN ($n = 468$)	GCD ($n = 2,862$)	Source Avg (Headline)	Pooled Test ($n = 3,330$)
ResNet-18	11.3M	60.90%	88.26%	74.58%	84.41%
ResNet-34	21.4M	60.68%	88.36%	74.52%	84.47%
ResNet-50	24.0M	60.26%	89.06%	74.66%	85.02%

Table 3: Primary Benchmark Results on the Held-Out Test Set ($N = 3,330$ images). The headline metric is the unweighted arithmetic mean of component point estimates (Source-Balanced Average), preventing GCD sample volume ($n = 2,862$) from dominating regional CCSN evaluation ($n = 468$). Bootstrap 95% confidence intervals quantify sampling variability for component and pooled holdouts in Appendix Table 5; no confidence interval is computed for the unweighted source-balanced mean, and no formal statistical equivalence is asserted for the 0.08 pp margin.

merely **0.08 percentage points**. Moreover, in balanced accuracy (macro-recall), ResNet-18 attains **73.77%**, slightly exceeding ResNet-50 (**73.44%**), and in macro-F1 ResNet-18 reaches **74.03%** versus **73.92%**. Because ResNet-18 requires 53% fewer parameters (11.3M vs. 24.0M) and 56% fewer FLOPs (1.82 vs. 4.12 GFLOPs), ResNet-18 is framed as the practical selected baseline for operational sky-imaging deployment, while avoiding formal claims of statistical equivalence.

- Primary vs. Secondary Metrics:** On the pooled test set ($n = 3,330$), ResNet-50 attains 85.02% versus 84.41% on ResNet-18. However, because GCD contributes 85.95% of the pooled test set (2,862 out of 3,330 images), pooled accuracy heavily reflects GCD whole-sky performance. Source-balanced averaging treats both observational modalities with equal weight, providing a more balanced evaluation of cross-source capability.
- Performance on Component Test Sets:** On the regional CCSN test component ($n = 468$), ResNet-18 attains the highest top-1 accuracy (60.90%), while ResNet-50 achieves higher accuracy on the whole-sky GCD component (89.06%).

6.3 Cross-Source Domain Discrepancy and Transfer Asymmetry

Ground-based sky imaging archives collected under differing operational objectives exhibit substantial domain shift across camera optics, sensor perspectives, and different class frequencies in each dataset. In the whole-sky GCD archive, Cumulonimbus represents 40.3% of images (5,764 samples), whereas in regional CCSN views, Cumulonimbus accounts for 22.1% (516 samples). Conversely, CCSN is dominated by Stratocumulus (31.2%) and features twice the relative frequency of Altocumulus (20.8% vs. 10.3% in GCD). Furthermore, GCD contains over six times as many images as CCSN (14,306 vs. 2,337), leading to substantial representation imbalance without active intervention. This empirical reality motivated the inverse-volume batch sampling weights ($w_i = 0.5/N_{\text{domain}(i)}$) defined in Section 5, providing balanced sampling across datasets.

These different class frequencies and camera views are consistent with the pronounced transfer asymmetry observed when single-source models are evaluated across sensor domains (Figure 2 and Appendix Table 5):

- CCSN $\rightarrow$ GCD (Regional to Whole-Sky):** Attains 57.86% overall accuracy and 57.94% balanced accuracy on ResNet-18. Models trained on regional cloud textures retain moderate sensitivity across whole-sky scenes, achieving 75.9% recall on GCD Cumulonimbus.
- GCD $\rightarrow$ CCSN (Whole-Sky to Regional):** Attains only 37.18% overall accuracy and 44.18% balanced accuracy. In particular, the GCD-trained model evaluated on CCSN achieves only **19.2%** Cumulonimbus recall, reflecting a substantial directional disparity compared to

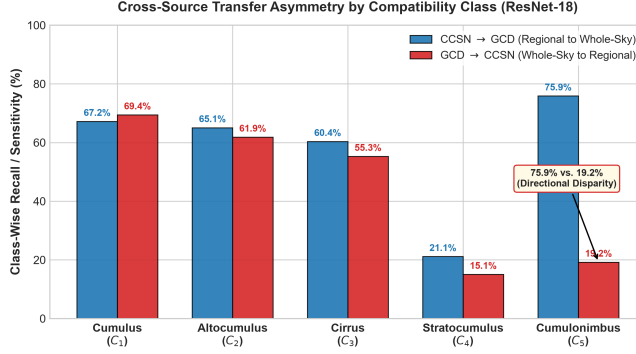


Figure 2: Cross-Source Transfer Asymmetry: Per-Class Sensitivity Disparity under Direct Cross-Source Transfer Evaluation for ResNet-18. Regional CCSN models retain moderate sensitivity across whole-sky scenes (75.9% Cumulonimbus recall on GCD), whereas whole-sky GCD models exhibit sharp directional sensitivity disparities on regional frames (19.2% Cumulonimbus recall on CCSN).

the 75.9% achieved in the reverse transfer direction. Models trained on whole-sky imagery are hypothesized to rely on the overall whole-sky layout, which is absent in localized telephoto frames.

While camera optics, field-of-view, and class distributions are known observational differences between the archives, attributing transfer differences to specific visual features remains an observational hypothesis rather than an isolated causal claim. Detailed per-condition metrics across all 24 configurations are compiled in Appendix Table 5.

7 Error Analysis and CCSN Generalization Gain

7.1 Substantiating the +4.06 Percentage Point Gain on CCSN

As recorded in the comprehensive results (Appendix Table 5), joint training with balanced batch sampling improves CCSN test component accuracy from 56.84% (CCSN-only baseline) to **60.90%** on ResNet-18, delivering a gain of **+4.06 percentage points**. Gains of +2.77 pp and +1.07 pp are similarly observed for ResNet-34 and ResNet-50.

Figure 3 provides normalized confusion matrices comparing the CCSN in-domain model against the joint model evaluated on the CCSN test component (468 images).

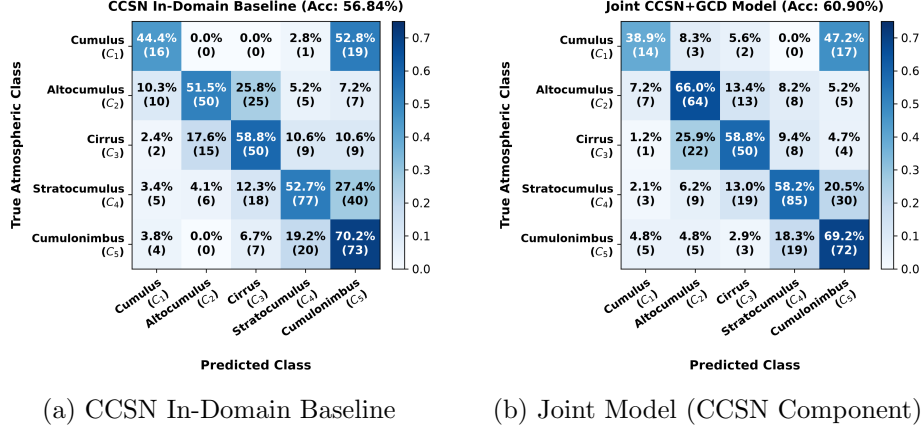


Figure 3: Normalized Confusion Matrices for ResNet-18 on the CCSN test component (468 images), showing classification distribution shifts before and after joint training with balanced batch sampling.

Comparison of Figures 3a and 3b shows how class-level decisions shift after joint training:

1. **Altocumulus Recognition:** Altocumulus (C_2) recall improves substantially from 51.5% to **66.0%** (+14.5 percentage points), with reduced confusion into Cirrus (from 25.8% down to 13.4%, while confusion into Stratocumulus shifts from 5.2% to 8.2%).
2. **Stratocumulus Stability:** Stratocumulus (C_4) recall increases from 52.7% to **58.2%**, consistent with improved discrimination between continuous low stratiform decks and convective rolls.
3. **High-Altitude Cirrus:** Cirrus (C_3) recall remains stable at **58.8%**, while Cumulonimbus (C_5) attains **69.2%**.

7.2 Persistent Failure Modes

Despite the improvements afforded by joint training, key meteorological ambiguities remain:

1. **Transitional Genera Ambiguity:** Altocumulus (C_2) and Stratocumulus (C_4) show persistent mutual misclassification due to shared morphological cloud roll textures during cloud layering transitions.
2. **Overcast Stratocumulus vs. Cumulonimbus:** Dense Stratocumulus decks cast dark grey diffuse lighting across ground sensors, visually resembling dark precipitation-bearing cloud bases (C_5) in single static visual frames without temporal barometric or radar data.
3. **Peripheral Horizon Clutter:** Wide-angle sky imagery contains peripheral horizon clutter (vegetation, masts, local topography) that presents non-meteorological visual features absent in regional sky views.

8 Conclusion and Future Work

In this work, we studied deep residual networks for cross-source ground-based cloud classification. By identifying cross-split duplicate exposures in public releases and applying Stratified Group K -Fold, we set up benchmark splits that prevent duplicate leakage. We defined five shared cloud classes to combine regional digital camera and wide-angle whole-sky datasets, applied balanced batch sampling, and evaluated convergence across hyperparameter configurations. Our results show that joint training achieves a source-balanced accuracy of **74.58%** on ResNet-18, within 0.08 percentage points of ResNet-50 (**74.66%**) while requiring 53% fewer parameters and 56% fewer FLOPs. Moreover, joint training improves classification accuracy on the sparse CCSN test component by +4.06 percentage points on ResNet-18, while cross-source evaluation highlights

pronounced transfer asymmetry between sensor modalities. These results support ResNet-18 as an effective, computationally lightweight baseline for ground-based cloud classification.

Future Work With 11.3M parameters and 1.82 GFLOPs, ResNet-18 is well suited for resource-constrained meteorological stations and mobile sensing platforms. Future work should evaluate on-device latency, runtime memory consumption, and INT8 quantization on embedded field hardware.

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Appendix A: Hyperparameter Exploration on Development Data

Table 4 documents empirical validation performance across the ten hyperparameter exploration configurations evaluated on the internal development validation set ($N = 2,663$ images). Sweeps explored differential learning rates, weight decay, label smoothing, and dropout regularization under AdamW and SGD with momentum. Trial 08 was selected as the production baseline configuration.

Trial ID	Configuration Description	Optimizer	Backbone LR	Head LR	ResNet-18	ResNet-34	ResNet-50
Trial 01	Baseline Differential LRs	AdamW	5×10^{-5}	5×10^{-4}	84.3%	83.7%	84.9%
Trial 02	Conservative (Low LR)	AdamW	2×10^{-5}	2×10^{-4}	83.3%	82.6%	83.1%
Trial 03	Aggressive (High LR)	AdamW	1×10^{-4}	1×10^{-3}	84.4%	85.0%	85.9%
Trial 04	Standard Momentum	SGD	1×10^{-3}	1×10^{-2}	82.2%	83.2%	83.9%
Trial 05	Conservative Momentum	SGD	5×10^{-4}	5×10^{-3}	81.4%	82.0%	82.4%
Trial 06	High Weight Decay (5×10^{-2})	AdamW	5×10^{-5}	5×10^{-4}	84.6%	84.9%	83.8%
Trial 07	Low Weight Decay (1×10^{-3})	AdamW	5×10^{-5}	5×10^{-4}	83.9%	83.7%	84.8%
Trial 08	Standard LRs (LS 0.0)	AdamW	5×10^{-5}	5×10^{-4}	83.7%	85.2%	85.1%
Trial 09	Heavy Dropout (0.4/0.5)	AdamW	5×10^{-5}	5×10^{-4}	85.4%	84.8%	85.7%
Trial 10	Fine Differential ($4 \times 10^{-5}/4 \times 10^{-4}$)	AdamW	4×10^{-5}	4×10^{-4}	84.5%	84.1%	85.1%

Table 4: Empirical validation accuracy across 10 hyperparameter exploration configurations on the internal development validation set (2,663 images). Displayed values reflect 5-epoch screening runs scored on unsmoothed validation cross-entropy loss.

Appendix B: Comprehensive Production Benchmark Reference

Table 5 reports the authoritative 24-condition experimental benchmark across all architectures, single-source in-domain baselines, cross-source transfer evaluations, and joint models on the held-out test set ($N = 3,330$ images), alongside exact 95% non-parametric percentile bootstrap confidence intervals ($B = 1,000$).

Model Architecture	Experimental Condition	Evaluation Set	Top-1 Acc (%)	Balanced Acc (%)	Macro-F1 (%)
ResNet-18 (11.3M)	1. CCSN In-Domain Model	CCSN Test Component	56.84 [52.35, 61.32]	55.55 [50.25, 60.60]	55.01 [49.72, 60.00]
	2. Cross-Source CCSN $\rightarrow$ GCD	GCD Test Component	57.86 [56.15, 59.57]	57.94 [55.95, 59.89]	59.33 [57.48, 61.09]
	3. GCD In-Domain Model	GCD Test Component	89.34 [88.29, 90.36]	92.00 [91.02, 92.92]	91.33 [90.33, 92.22]
	4. Cross-Source GCD $\rightarrow$ CCSN	CCSN Test Component	37.18 [32.90, 41.67]	44.18 [39.48, 48.94]	36.37 [31.89, 40.63]
	5. Joint Model (CCSN+GCD)	CCSN Test Component	60.90 [56.20, 65.17]	58.23 [53.17, 63.52]	58.11 [52.78, 63.10]
	6. Joint Model (CCSN+GCD)	GCD Test Component	88.26 [87.04, 89.45]	89.32 [88.07, 90.50]	89.95 [88.74, 91.06]
	7. Joint Model (CCSN+GCD)	Pooled Test Set	84.41 [83.21, 85.71]	84.70 [83.34, 86.03]	85.17 [83.87, 86.42]
	8. Joint Model (CCSN+GCD)	Source-Balanced Avg	74.58	73.77	74.03
ResNet-34 (21.4M)	1. CCSN In-Domain Model	CCSN Test Component	57.91 [53.42, 62.18]	52.96 [48.22, 57.86]	53.49 [48.30, 58.33]
	2. Cross-Source CCSN $\rightarrow$ GCD	GCD Test Component	30.68 [28.90, 32.42]	38.53 [36.64, 40.39]	33.74 [31.65, 35.68]
	3. GCD In-Domain Model	GCD Test Component	89.06 [87.91, 90.15]	91.35 [90.25, 92.33]	90.80 [89.69, 91.77]
	4. Cross-Source GCD $\rightarrow$ CCSN	CCSN Test Component	34.83 [30.56, 39.10]	42.37 [38.28, 46.69]	33.10 [29.03, 37.27]
	5. Joint Model (CCSN+GCD)	CCSN Test Component	60.68 [56.20, 64.96]	57.86 [53.09, 62.56]	57.64 [52.81, 62.19]
	6. Joint Model (CCSN+GCD)	GCD Test Component	88.36 [87.18, 89.55]	89.87 [88.68, 91.01]	90.07 [88.96, 91.11]
	7. Joint Model (CCSN+GCD)	Pooled Test Set	84.47 [83.27, 85.74]	84.99 [83.73, 86.35]	85.20 [84.00, 86.50]
	8. Joint Model (CCSN+GCD)	Source-Balanced Avg	74.52	73.87	73.85
ResNet-50 (24.0M)	1. CCSN In-Domain Model	CCSN Test Component	59.19 [54.91, 63.68]	55.48 [50.77, 60.84]	55.64 [50.85, 60.67]
	2. Cross-Source CCSN $\rightarrow$ GCD	GCD Test Component	43.99 [42.17, 45.81]	52.85 [50.89, 54.82]	48.38 [46.36, 50.37]
	3. GCD In-Domain Model	GCD Test Component	89.45 [88.29, 90.57]	90.99 [89.88, 91.99]	91.22 [90.16, 92.16]
	4. Cross-Source GCD $\rightarrow$ CCSN	CCSN Test Component	33.76 [29.70, 38.25]	44.19 [40.30, 48.29]	31.97 [27.90, 36.04]
	5. Joint Model (CCSN+GCD)	CCSN Test Component	60.26 [55.77, 64.32]	56.18 [51.33, 60.90]	56.73 [51.33, 61.28]
	6. Joint Model (CCSN+GCD)	GCD Test Component	89.06 [87.84, 90.22]	90.71 [89.61, 91.82]	91.11 [90.08, 92.14]
	7. Joint Model (CCSN+GCD)	Pooled Test Set	85.02 [83.78, 86.25]	85.46 [84.19, 86.79]	86.03 [84.86, 87.25]
	8. Joint Model (CCSN+GCD)	Source-Balanced Avg	74.66	73.44	73.92

Table 5: Comprehensive Benchmark Reference Table across all 24 Experimental Conditions on the Held-Out Test Set (15-Epoch Production Runs with 95% Bootstrap Confidence Intervals).